Final Technical Report

Ishaan Mehrotra

Department of Biomedical Engineering, Duke University

BME290L: Medical Technology Prototyping Skills

Dr. Mark Palmeri, M.D. Ph.D.

29 September 2023

Contents

1 Needs/Specifications……3

1.1 Enclosure……3

1.2 Electronics……..4

2 Design Documents……5

2.1 Functional Decomposition……5

2.2 ECAD

2.2.1 Schematic Capture……6

2.2.2 PCB Layout……...8

2.2.3 Bill of Materials……...13

2.3 CAD……14

2.4 Images of Enclosure……19

3 Testing and Analysis……...22

3.1 Testing to Specification……22

3.2 Test Power Consumption……28

3.3 Optimizations……...29

4 Microcontroller……30

1 Needs/Specifications

1.1 Enclosure

For the enclosure, needs and specifications include:

A) Able to be ergonomically held in one hand.

B) Button(s) should be easily accessible to press while hand is holding device.

C) Stable operation on a flat surface without lifting up.

D) Simple and easy battery replacement.

E) Able to prevent accidental power switch off during use.

F) Accessible potentiometer knobs that can be used by second hand while in operation.

G) Securely mounting of board with four mounting holes.

H) Accessible test pads for electronics testing without removing the entire board.

I) Internally routed and secured wires between PCB and components.

J) Securely mounted all peripheral components within the enclosure.

K) Securely hold batteries using 2x2 or 1x4 AA battery holders.

L) Smallest size possible (while attaining to all other constraints) is preferred

M) A lighter weight is desirable

N) Balanced weight allowing for comfortable feel when held

O) Prioritization of ergonomics and aesthetics avoiding corners and straight edges

P) Minimization of stress concentrations to ensure durability

1.2 Electronics

For the electronics involved, needs and specifications include:

A) **Power Activation**: Turning on the switch establishes battery connection to the circuit; turning it off blocks the power and thus no connection transpires.

B) **Blink Trigger**: Pressing the button activates a monostable output that initiates the blinking of the LED.

C) **Potentiometer Control**: Two potentiometers are utilized to manage the LED blinking frequency and duration of blinking.

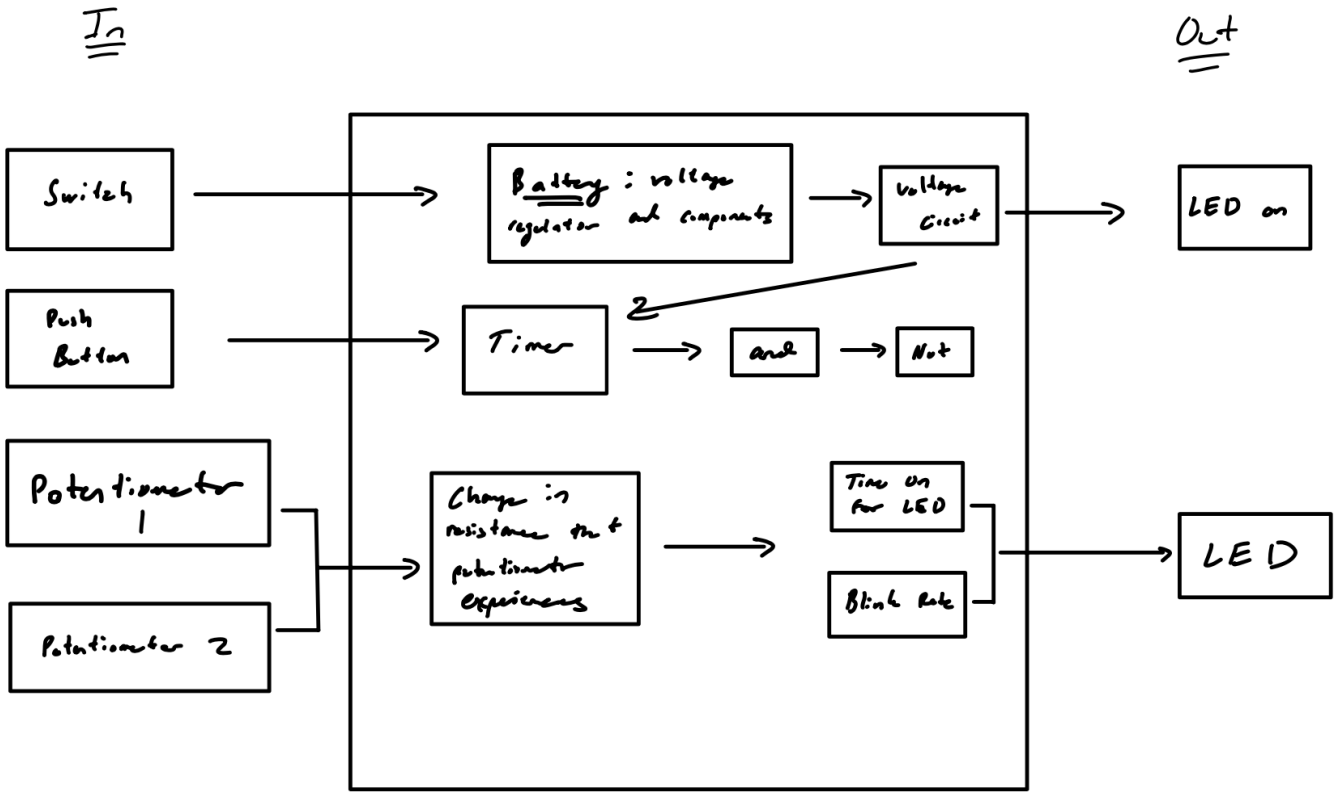
D) **Blinking Activation**: Upon pressing the button, the LED must blink in a specified pattern.

E) **Frequency Control**: LED blinking frequency adjustable within the range of 5 Hz to 10 Hz, based on the astable potentiometer setting.

F) **Duration Adjustment**: Duration of LED blinking adjustable between 3 to 10 seconds depending on the monostable potentiometer setting.

2 Design Documents

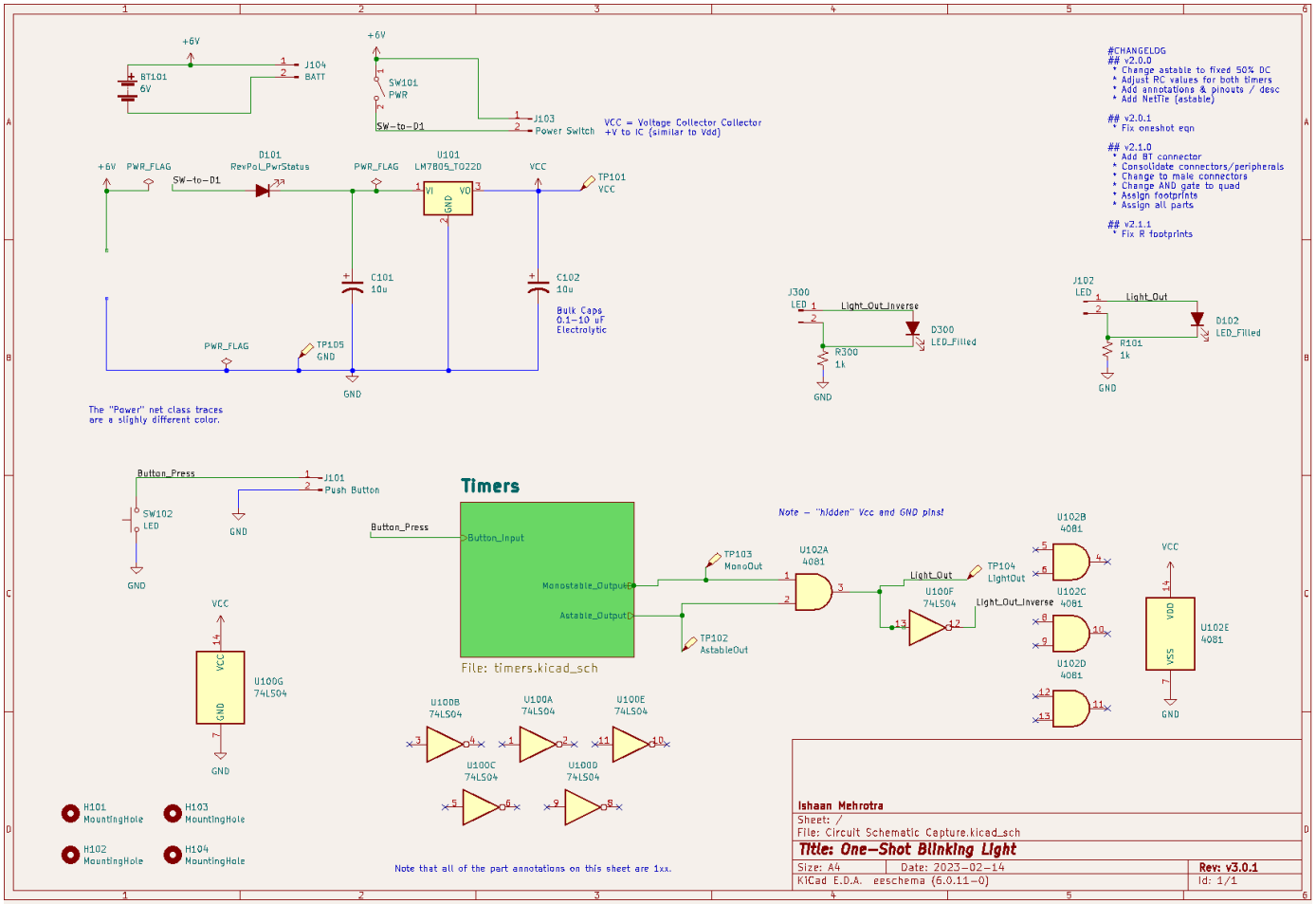
2.1 Functional Decomposition

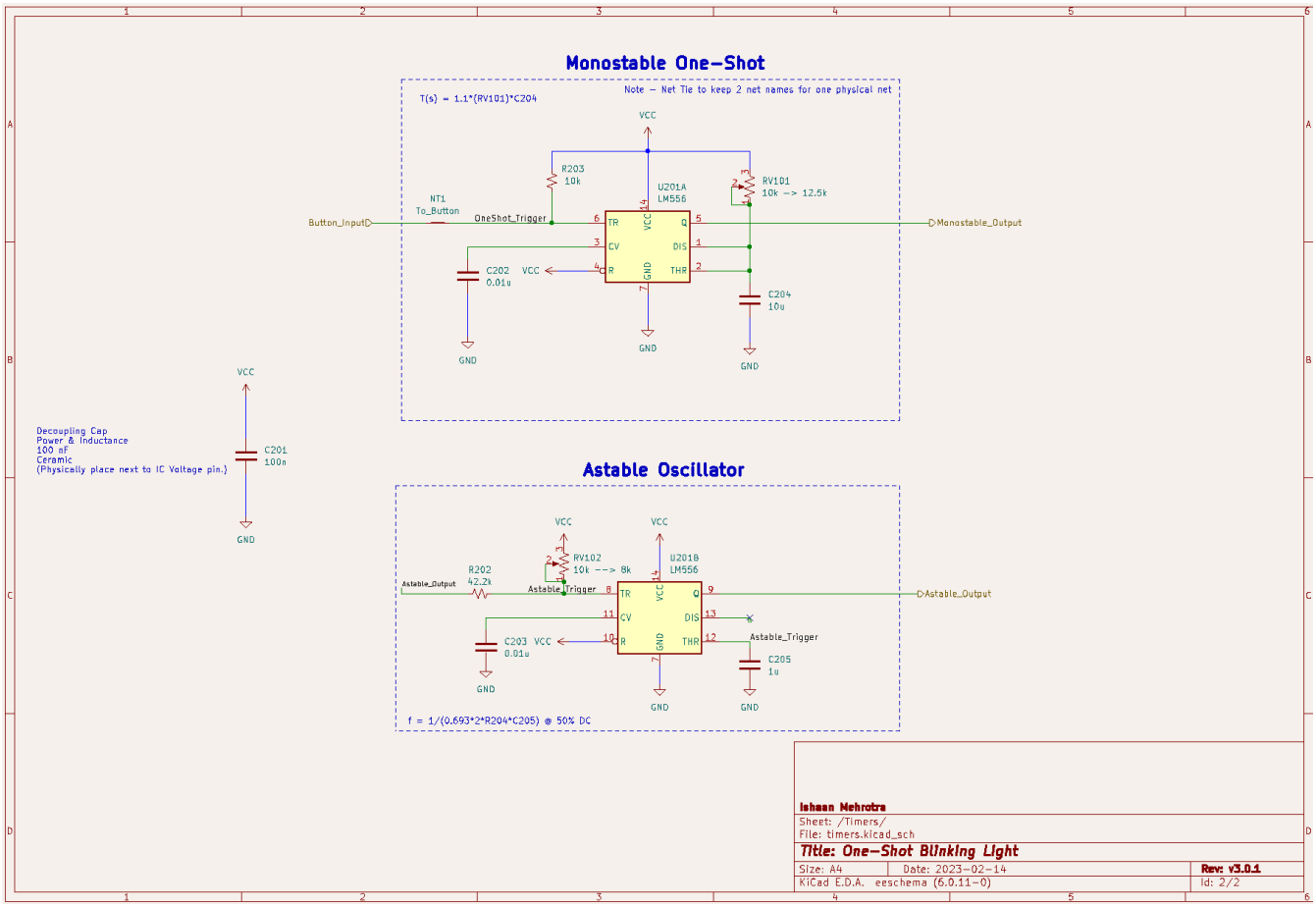


only had 1 LED in my system

2.2 ECAD

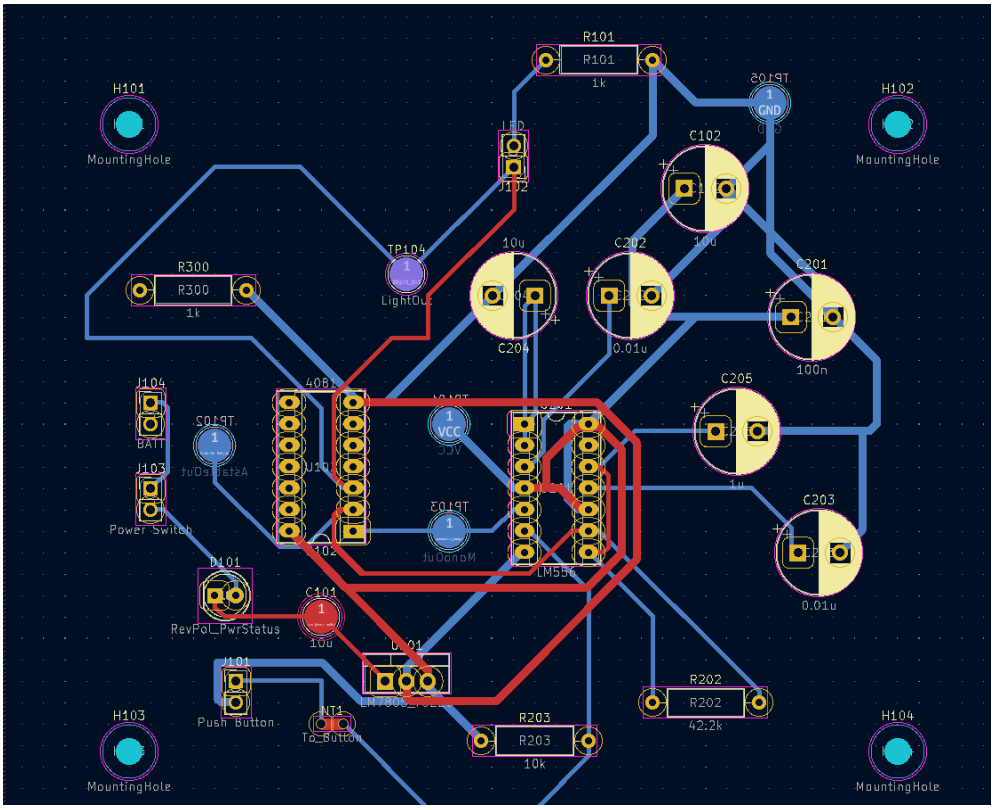
2.2.1 Schematic Capture

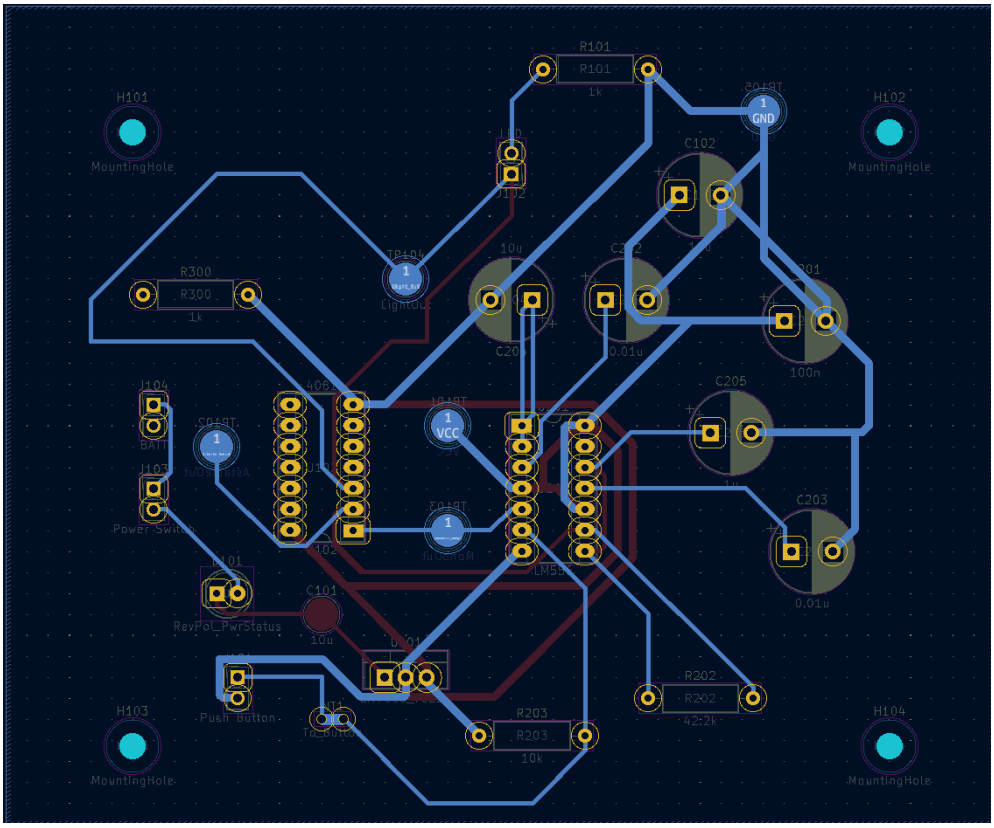


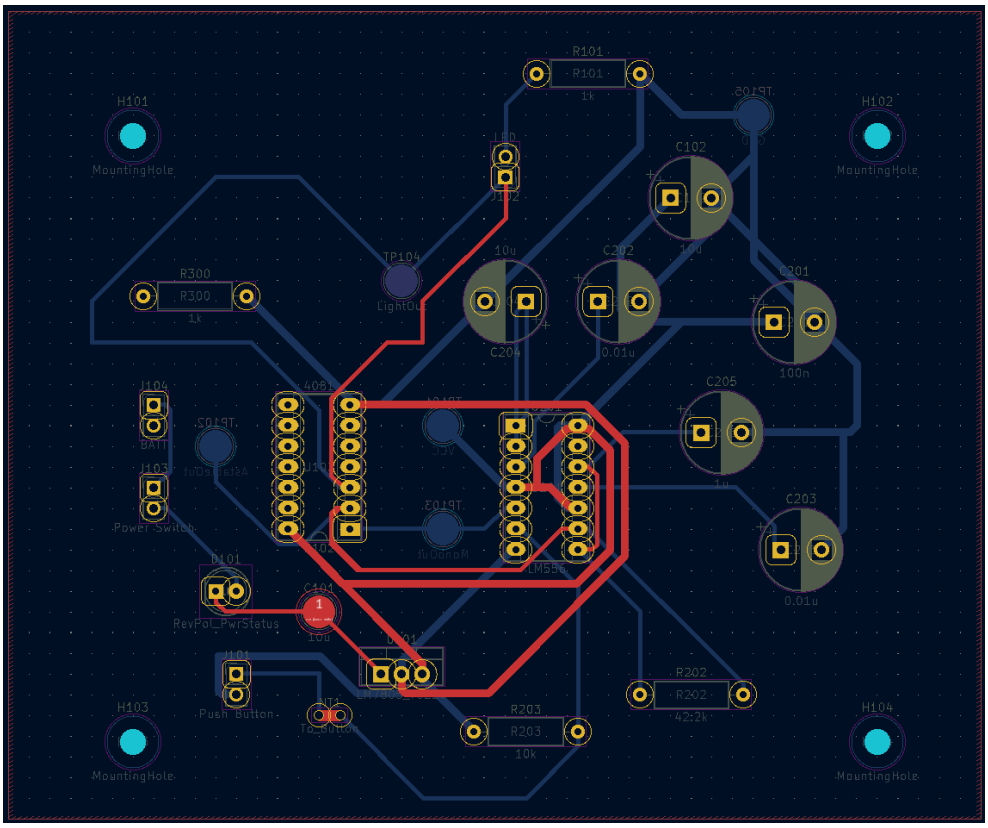

Monostable One-Shot
T(s) = 1.1*(RV101)*C204
Note - Net Tie to keep 2 net names for one physical net
VCC
R203
10k
RV101
10k --> 12.5k
U201A
LM556
NT1
To_Button
Button_Input
OneShot_Trigger
Monostable_Output
TR
CV
R
Q
DIS
THR
GND
VCC
C202
0.01u
C204
10u
Decoupling Cap
Power & Inductance
100 nF
Ceramic
(Physically place next to IC Voltage pin.)
C201
100n
Astable Oscillator
RV102
10k --> 8k
R202
42.2k
U201B
LM556
Astable_Output
Astable_Trigger
C203
0.01u
C205
1u
f = 1/(0.693*2*R204*C205) @ 50% DC
Ishaan Mehrotra
Sheet: /Timers/
File: timers.kicad_sch
Title: One-Shot Blinking Light
Size: A4
Date: 2023-02-14
Rev: v3.0.1
KiCad E.D.A. eeschema (6.0.11-0)
Id: 2/2

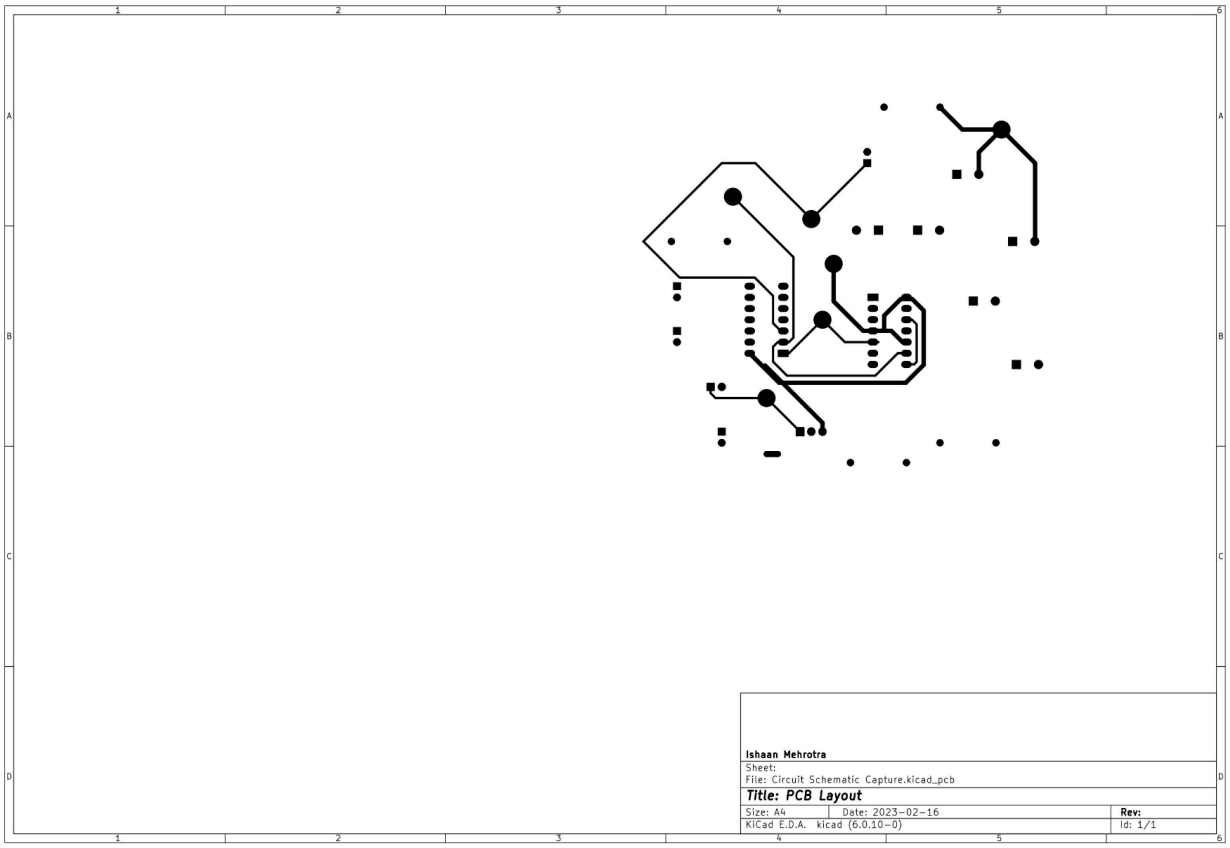
2.2 ECAD

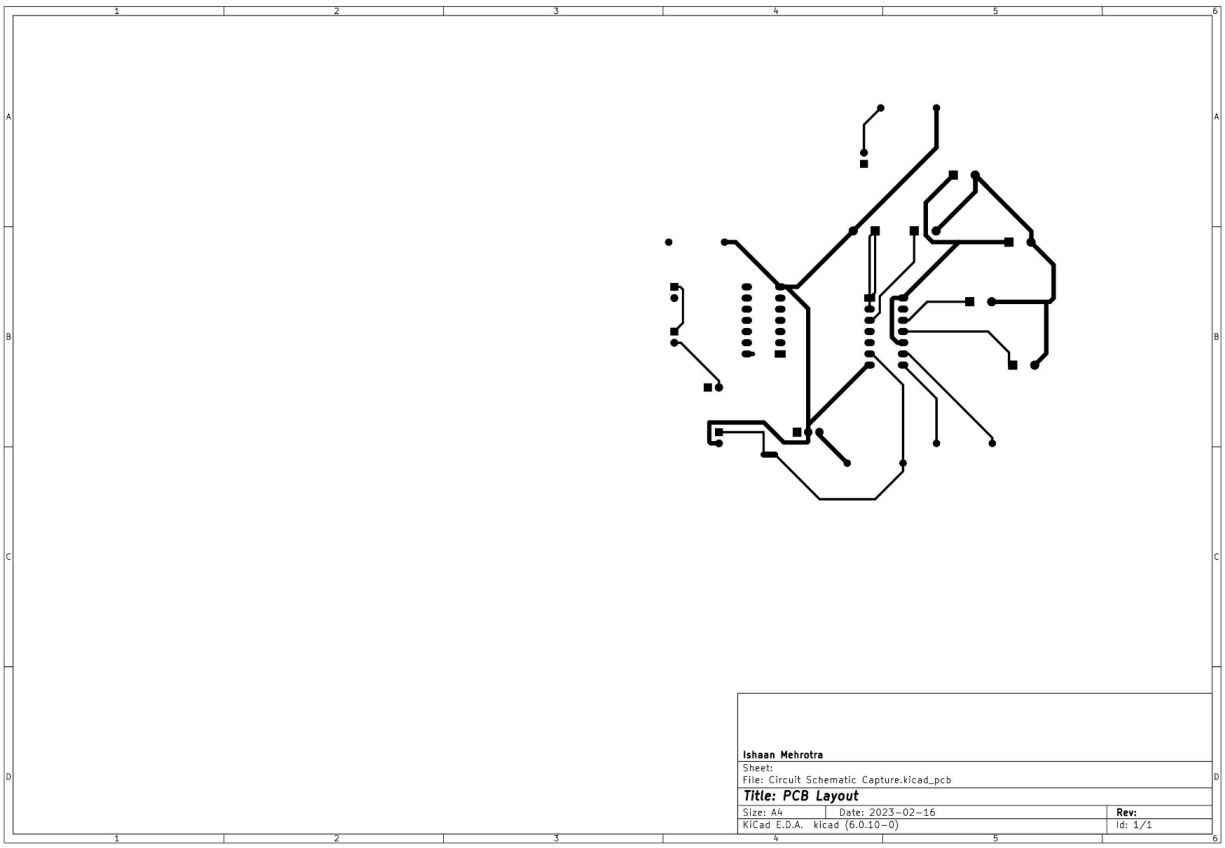
2.2.2 PCB Layout

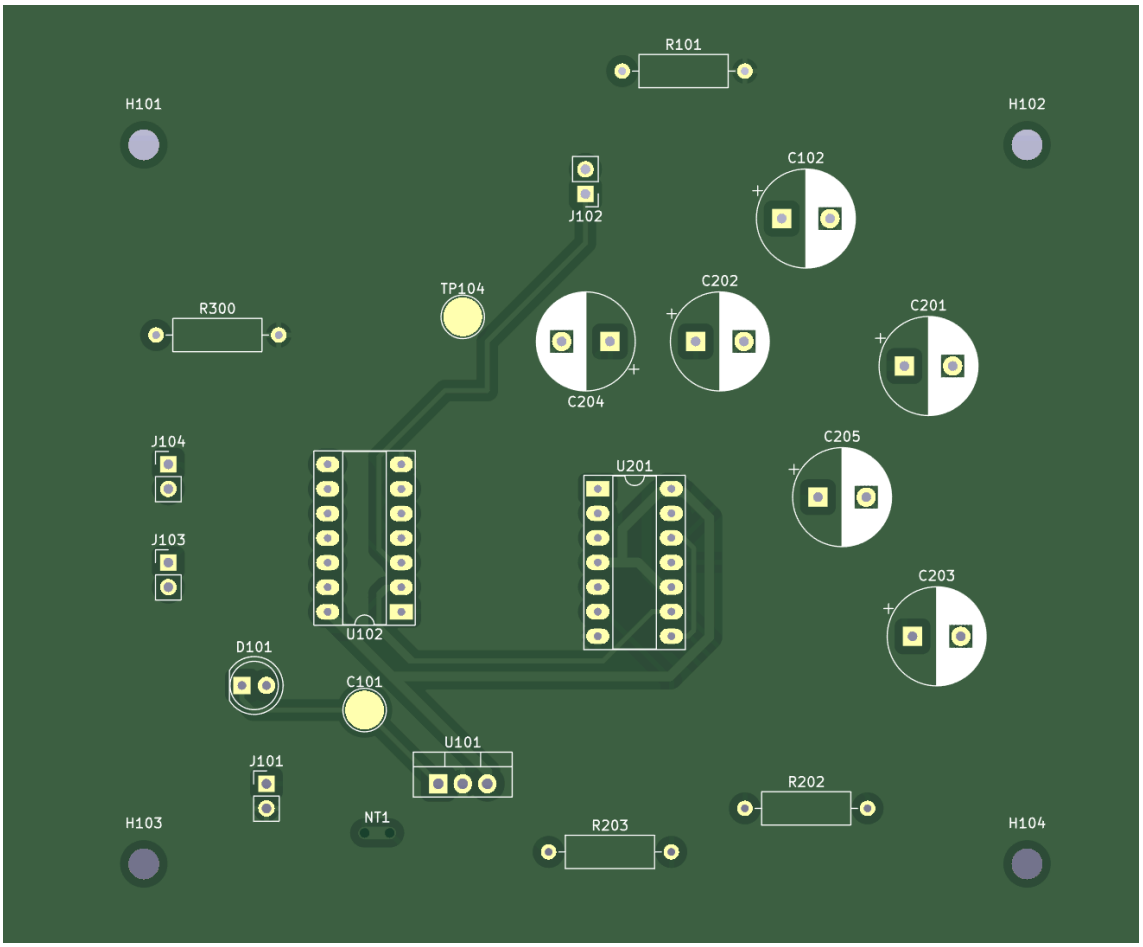


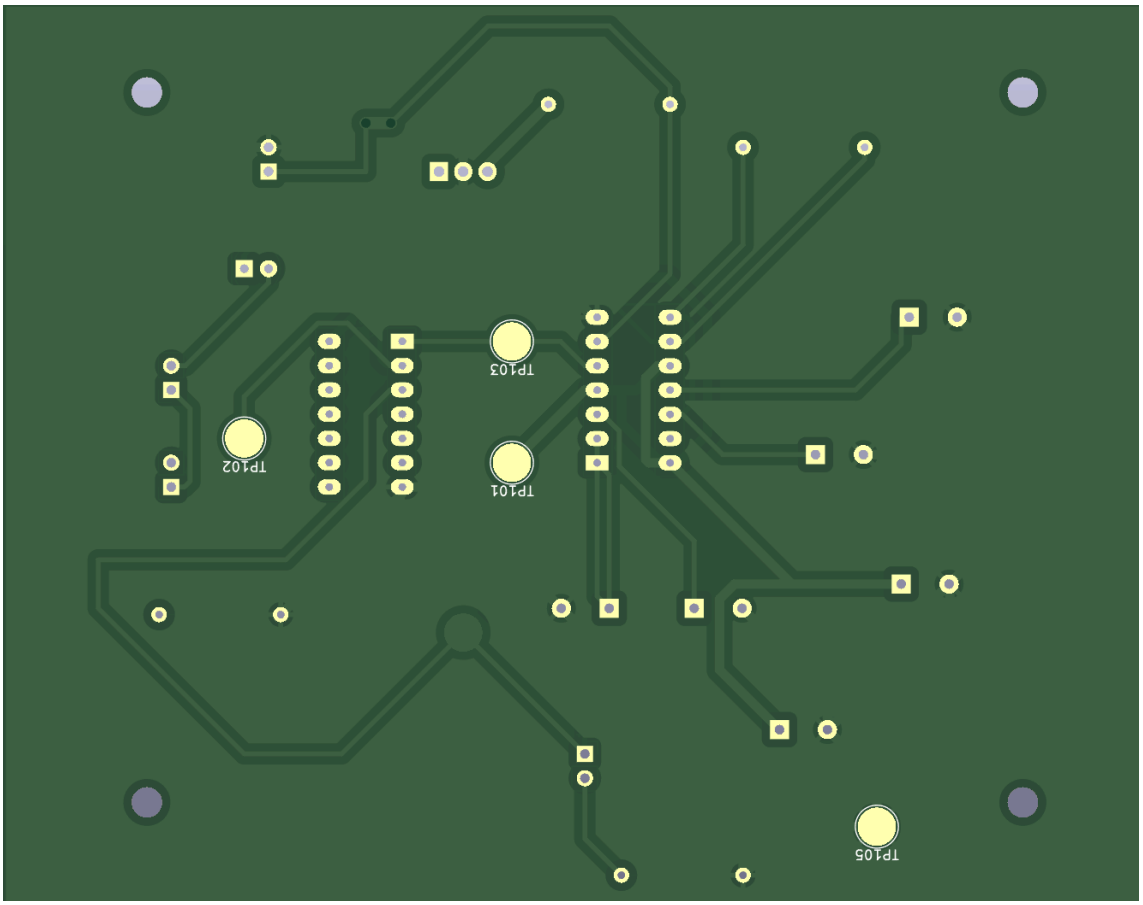



R101
R101
1k
H101
MountingHole
H102
MountingHole
C102
100u
TP104
LightOut
R300
R300
1k
10u
C204
C202
0.01u
C201
100n
J104
BATT
J103
Power Switch
C205
1u
C203
0.01u
D101
RevPol_PwrStatus
C101
10u
J101
Push Button
R202
R202
42.2k
R203
R203
10k
H103
MountingHole
H104
MountingHole


Ishaan Mehrotra
Sheet:
File: Circuit Schematic Capture.kicad_pcb
Title: PCB Layout
Size: A4
Date: 2023-02-16
Rev:
KiCad E.D.A. kicad (6.0.10-0)
Id: 1/1


Ishaan Mehrotra
Sheet:
File: Circuit Schematic Capture.kicad_pcb
Title: PCB Layout
Size: A4
Date: 2023-02-16
Rev:
KiCad E.D.A. kicad (6.0.10-0)
Id: 1/1


R101
H101
H102
C102
J102
TP104
R300
C202
C201
C204
J104
C205
U201
J103
U102
C203
D101
C101
U101
J101
R202
H103
NT1
R203
H104


TP103
TP102
TP101
TP105

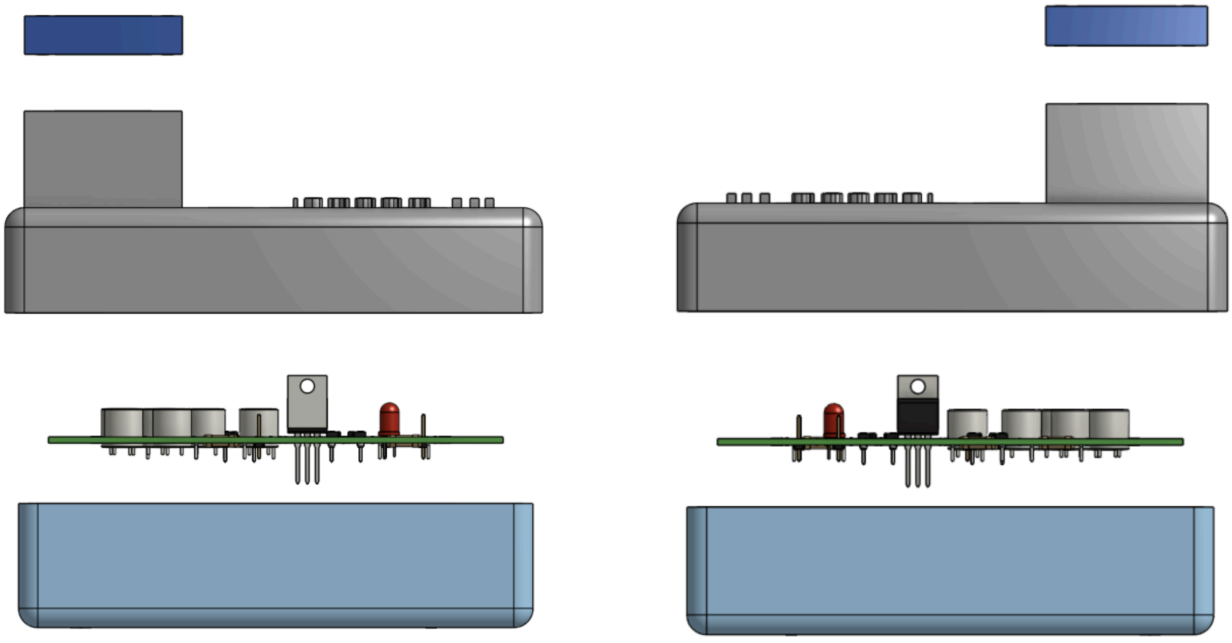
2.2 ECAD

2.2.3 Bill of Materials

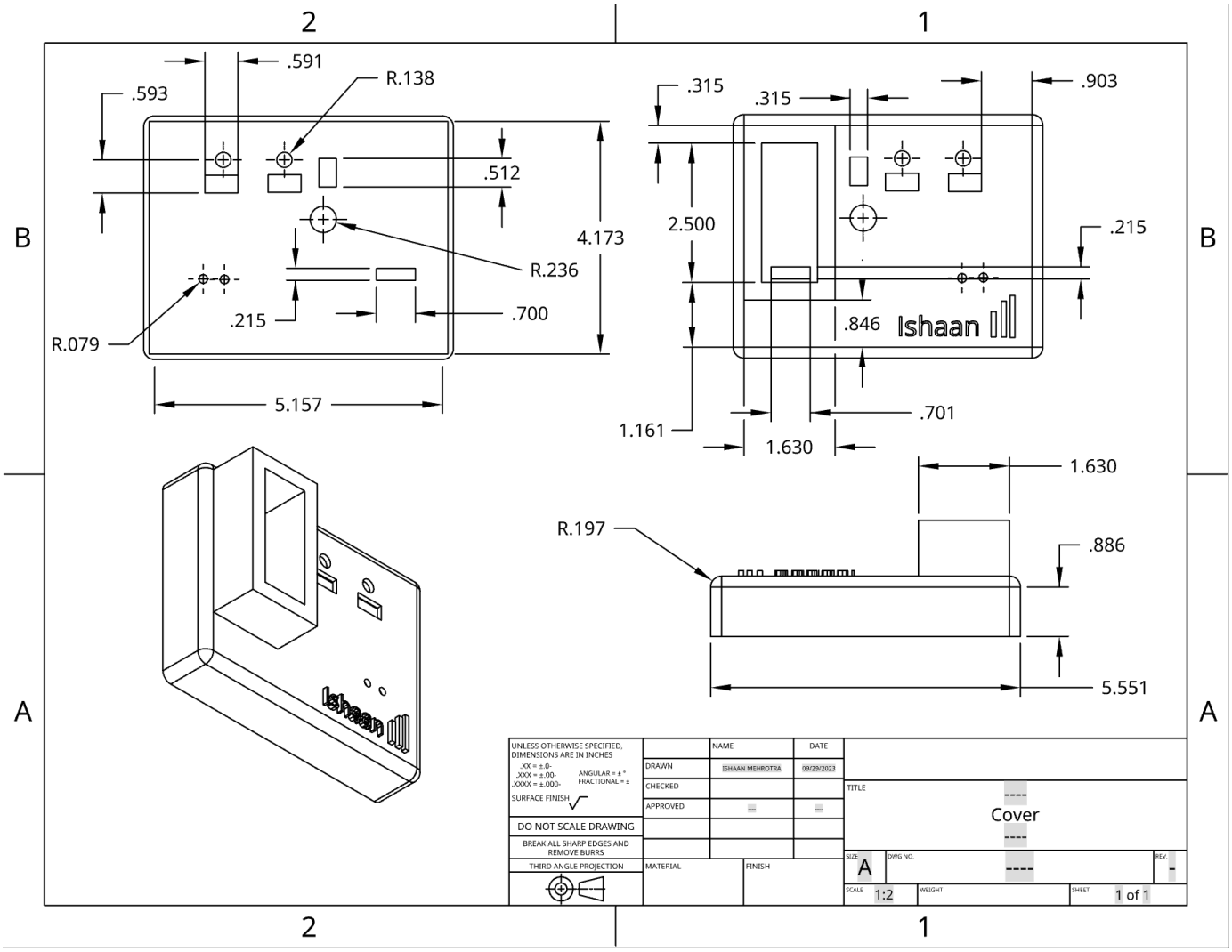
□□□	Sourced	Placed	References	Value	Footprint	Quantity
1	☐	☐	C102, C204	10u	CP_Radial_D10.0mm_P5.00mm	2
2	☐	☐	C202, C203	0.01u	CP_Radial_D10.0mm_P5.00mm	2
3	☐	☐	C101	10u	TestPoint_Pad_D4.0mm	1
4	☐	☐	C201	100n	CP_Radial_D10.0mm_P5.00mm	1
5	☐	☐	C205	1u	CP_Radial_D10.0mm_P5.00mm	1
6	☐	☐	R101, R300	1k	R_Axial_DIN0309_L9.0mm_D3.2mm_P12.70mm_Horizontal	2
7	☐	☐	R202	42.2k	R_Axial_DIN0309_L9.0mm_D3.2mm_P12.70mm_Horizontal	1
8	☐	☐	R203	10k	R_Axial_DIN0309_L9.0mm_D3.2mm_P12.70mm_Horizontal	1
9	☐	☐	D101	RevPol_PwrStatus	LED_D5.0mm	1
10	☐	☐	U101	LM7805_TO220	TO-220-3_Vertical	1
11	☐	☐	U102	4081	DIP-14_W7.62mm_Socket_LongPads	1
12	☐	☐	U201	LM556	DIP-14_W7.62mm_Socket_LongPads	1
13	☐	☐	H101, H102, H103, H104	MountingHole	MountingHole_3.2mm_M3	4
14	☐	☐	TP101	VCC	TestPoint_Pad_D4.0mm	1
15	☐	☐	TP102	AstableOut	TestPoint_Pad_D4.0mm	1
16	☐	☐	TP103	MonoOut	TestPoint_Pad_D4.0mm	1
17	☐	☐	TP104	LightOut	TestPoint_Pad_D4.0mm	1
18	☐	☐	TP105	GND	TestPoint_Pad_D4.0mm	1
19	☐	☐	J101	Push Button	PinHeader_1x02_P2.54mm_Vertical	1
20	☐	☐	J102	LED	PinHeader_1x02_P2.54mm_Vertical	1
21	☐	☐	J103	Power Switch	PinHeader_1x02_P2.54mm_Vertical	1
22	☐	☐	J104	BATT	PinHeader_1x02_P2.54mm_Vertical	1

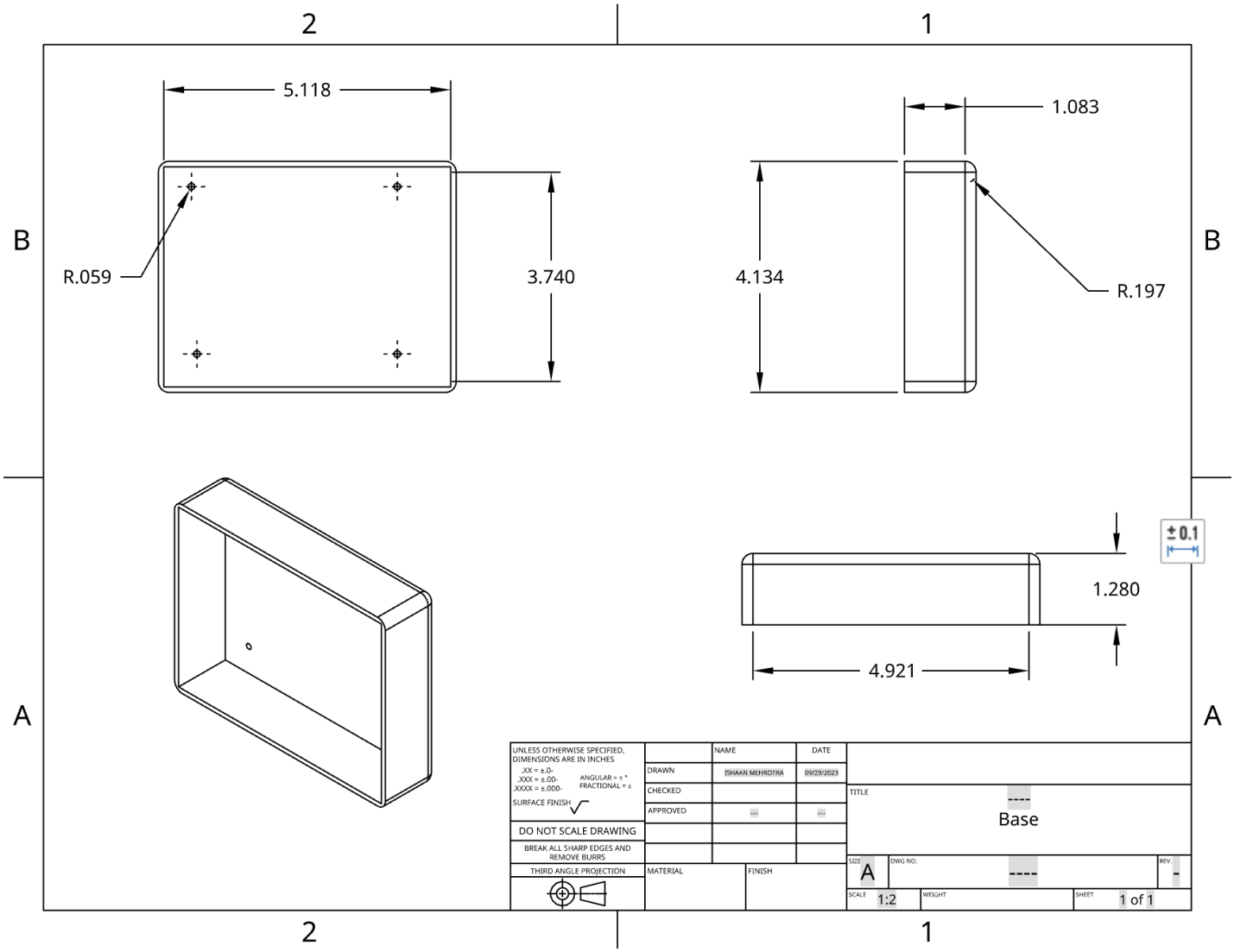
2.3 CAD

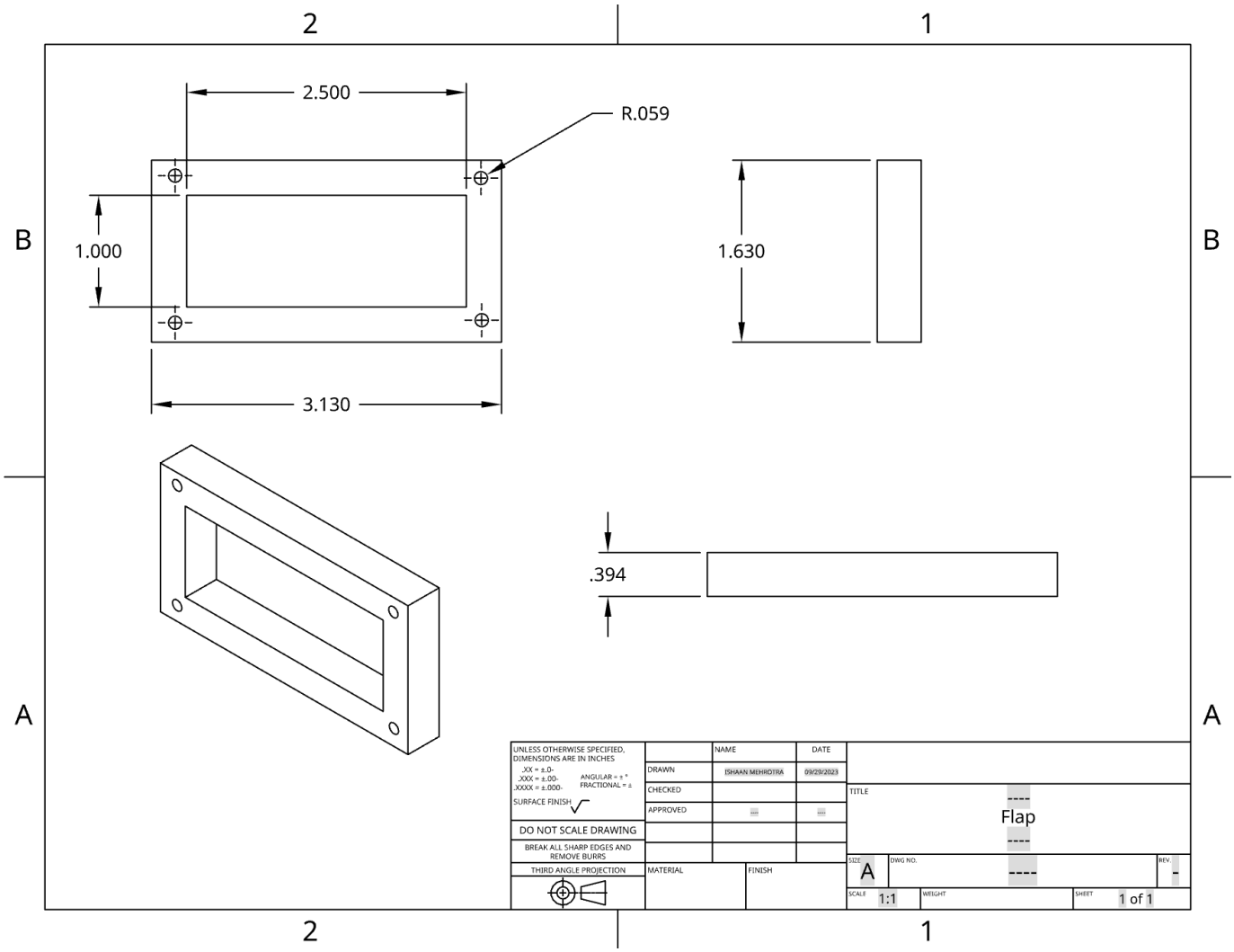
3D Renderings of the Exploded Assembly:



Orthographic Mechanical Drawings of All Parts:




2
1
B
A
5.118
3.740
R.059
1.083
4.134
R.197
1.280
4.921
±0.1
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
ISHAAN MEHROTRA
09/29/2023
CHECKED
APPROVED
MATERIAL
FINISH
TITLE
Base
SIZE
A
DWG NO.
REV.
SCALE
1:2
WEIGHT
SHEET
1 of 1


2
1
B
A
2.500
R.059
1.000
1.630
3.130
.394
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
ISHAAN MEHROTRA
09/29/2023
CHECKED
APPROVED
MATERIAL
FINISH
TITLE
Flap
SIZE
A
DWG NO.
REV.
SCALE
1:1
WEIGHT
SHEET
1 of 1

2.4 Images of Enclosure

3 Testing and Analysis

3.1 Testing to Specification

Enclosure Needs/Specifications

A) Able to be ergonomically held in one hand.

→ *Testing Strategy:* if able to hold in one hand, the test will pass; otherwise, fail.

→ **Result:** PASSED

B) Button(s) should be easily accessible to press while hand is holding device.

→ *Testing Strategy*: if able to press the button(s) while hand is holding the device, the test will pass; otherwise, fail.

→ **Result:** PASSED

C) Stable operation on a flat surface without lifting up.

→ *Testing Strategy:* if able to operate device on a flat surface without lifting up, the test will pass; otherwise, fail.

→ **Result:** PASSED

D) Simple and easy battery replacement.

→ *Testing Strategy:* if able to easily replace the batteries in the battery pack, the test will pass; otherwise, fail.

→ **Result:** PASSED

E) Able to prevent accidental power switch off during use.

→ *Testing Strategy:* if able to operate the device without accidentally turning off the switch, the test will pass.

→ **Result:** PASSED

F) Accessible potentiometer knobs that can be used by second hand while in operation.

→ *Testing Strategy:* if able to operate potentiometers with a second hand not holding device, the test will pass.

→ **Result:** PASSED

G) Securely mounting of board with four mounting holes.

→ *Testing Strategy:* if the board stays securely mounted to the enclosure while the enclosure is tossed up in the air and safely caught, the test will pass.

→ **Result:** PASSED

H) Accessible test pads for electronics testing without removing the entire board.

→ *Testing Strategy:* if testing points can be reached while the board is in enclosure, the test will pass.

→ **Result:** PASSED

I) Internally routed and secured wires between PCB and components.

→ *Testing Strategy:* if wires between the board and components inside the enclosure remain connected while the enclosure is rotated or moved from one spot to another, the test will pass.

→ **Result:** FAILED; the wires connecting the push button and power switch to the board kept coming undone from their respective connection points on the PCB as the connection point was a bit too small to hold it in place tightly; to fix this, I would make sure the connection pins are long enough to keep the wires securely attached to their points.

J) Securely mounted all peripheral components within the enclosure.

→ *Testing Strategy:* if the components within the enclosure stay in the enclosure when thrown up in the air or moved to a different spot on a table, the test will pass.

→ **Result:** PASSED

K) Securely hold batteries using 2x2 or 1x4 AA battery holders.

→ *Testing Strategy:* if the battery pack can hold the batteries and is accessible at all times, this test will pass.

→ **Result:** PASSED

L) Smallest size possible (while attaining to all other constraints) is preferred

→ Optimized as much as possible!

M) A lighter weight is desirable

→ Optimized as much as possible!

N) Balanced weight allowing for comfortable feel when held

→ *Testing Strategy:* if the weight feels balanced and comfortable in hand, the test will pass.

→ **Result:** PASSED

O) Prioritization of ergonomics and aesthetics avoiding corners and straight edges

→ *Testing Strategy:* if the enclosure does not have any sharp edges and has mainly rounded corners, the test will pass.

→ **Result:** PASSED

P) Minimization of stress concentrations to ensure durability

→ *Testing Strategy:* if no fatal stress concentrations occur when dropped from 6 ft, the test will pass.

→ **Result.** FAILED; when dropped from 6 ft, the board remained safe and the components remained in their positions in the enclosure; however, the cover of the

enclosure came undone from the base containing the board, leading to the fact that the board would stop working.

Electronics Needs/Specifications

A) **Power Activation**: Turning on the switch establishes battery connection to the circuit; turning it off blocks the power and thus no connection transpires.

→ *Testing Strategy*: if turning on the switch establishes the desired connection, this test will pass.

→ **Result:** PASSED

B) **Blink Trigger**: Pressing the button activates a monostable output that initiates the blinking of the LED.

→ *Testing Strategy:* if pressing the button leads to the blinking of the LED, the test will pass.

→ **Result:** FAILED

C) **Potentiometer Control**: Two potentiometers are utilized to manage the LED blinking frequency and duration of blinking.

→ *Testing Strategy:* if the board has two potentiometers that are used to manage LED blinking frequency and duration, the test will pass.

→ **Result:** PASSED

D) **Blinking Activation**: Upon pressing the button, the LED must blink in a specified pattern.

→ *Testing Strategy:* if LED blinks in pattern after pressing button, the test will pass.

→ **Result:** FAILED

E) **Frequency Control**: LED blinking frequency adjustable within the range of 5 Hz to 10 Hz, based on the astable potentiometer setting.

→ *Testing Strategy:* if LED frequency is adjustable within this range, the test will pass.

→ **Result**: FAILED

F) **Duration Adjustment**: Duration of LED blinking adjustable between 3 to 10 seconds depending on the monostable potentiometer setting.

→ *Testing Strategy:* if LED blinking is adjustable between this range, the test will pass.

→ **Result**: FAILED

The electronics testing above that failed were most likely due to the following reasons: in order for the circuit to work entirely, each component of the circuit needs to be connected correctly to one another in a manner that allows for it to work and run. In this case, although the PCB board was functioning, there may have been a mistake in the connections of the circuit that induced the LED to not blink in the desired manner. Additionally, my PCB board only incorporated one LED instead of the desired two. The LED's failure to blink within the 5-10 Hz frequency range as adjusted by the astable potentiometer, and the inability to control the blinking duration within 3-10 seconds using the monostable potentiometer, can be attributed to a range of potential issues within the circuit. These issues may include misconfigured potentiometer settings, component damage (very plausible due to the fact that these potentiometers were given last semester and have been through a lot), power supply instability, timing component discrepancies, electromagnetic interference, and schematic or PCB design errors. Careful examination and systematic troubleshooting of these factors should help identify and address the root cause of the problem, ensuring the LED functions as intended. In the future, to ensure the reliable blinking of the LED in my future circuit designs, I'll take a comprehensive approach. I'll begin by

conducting thorough testing and prototyping to validate the circuit's functionality, exploring a range of potentiometer settings to confirm that the LED operates within the specified frequency and duration ranges. Clear and detailed documentation of the circuit design, including component values and wiring diagrams, will be crucial for troubleshooting and reference. I'll prioritize the use of high-quality, reputable components to minimize the risk of component failures. I'll also consider having the circuit schematic reviewed by experienced engineers to identify and rectify design errors or areas for improvement. I'll pay careful attention to PCB layout, focusing on signal integrity, grounding, and noise reduction. Ensuring a stable power supply with proper voltage regulation and implementing electromagnetic interference (EMI) mitigation techniques will additionally be a part of my strategy. I'll select components with suitable tolerances and, if applicable, design user-friendly controls for potentiometers. Incorporating feedback mechanisms for users and conducting thorough quality assurance testing before distribution will be essential. Finally, I'll embrace a mindset of continuous improvement, gathering user feedback and conducting post-release testing to refine the design as needed. These steps will collectively enhance the reliability and performance of my future circuits.

3.2 Test Power Consumption

Even though there seemed to be connection throughout the board when the switch was turned on and connected to a power source, there was a shorting of the circuit that induced the LED to not blink; therefore, the device currently cannot be properly tested for power consumption or battery life. However, this testing will be done in the Microcontroller section of this report and results along with oscilloscope readings can be observed there.

3.3 Optimizations

The enclosure has been designed in a way to optimize size and weight, specifically in the manner of creating a system that will hold the PCB board (given its size) and the required components as well. There was enough room in the enclosure for the specific components to be added in; however, I would focus on optimizing where these components are placed and where their connections onto the board are located as well in order to build the most efficient and wire-tangle free system. Moreover, my design could be revised by making the board significantly smaller as there was a lot of space on the board that increased the size and weight of the overall enclosure as well as a result. If I did this, then I would have to make sure where the components are placed on the board when building the circuit in order to maximize solder area and be able to solder the connections correctly; if the components are close to each other, soldering is much more difficult. I should have originally made the PCB much smaller in order to decrease the size of the enclosure as it is not that much on the smaller side of contraptions. I did a good job of keeping space for the internal components of the enclosure; additionally, I would make sure that my last iteration of the PCB board was complete before beginning the creation of the enclosure. Finally, I would revise the length of the wires for the potentiometers and add potentiometer connections onto the PCB board instead of having to manually add direct connections of them on the back side of the board. This is where the nodes connected to each other to act as the potentiometers do in the drawing.

4 Microcontroller

```
// LOAD ALL LIBRARIES FIRST
#include <Arduino.h>
//#include <avr/sleep.h> // only works for the AVR-based boards; not the Nano 33 BLE

// DEFINE ALL PREPROCESSOR MACROS
// see pinout in datasheet to know what pins are where / do what
// NOTE: macros are defined in ALL CAPS to help identify their usage in the code (in
contrast to variables)
// All device specification-related values should be defined here, *not* in the
downstream code! You should
// not have to search the code for where values are defined / used.
#define LED_PIN 13 // D13
#define INV_LED_PIN 10 // D10
#define BUTTON_INTERRUPT_PIN 2 // D2
#define ASTABLE_POT_PIN A0
#define MONOSTABLE_POT_PIN A1
#define SLEEP_DELAY_MS 500
#define HIGH_VOLT 3.3 // microcontroller-specific logic voltage value
#define ANALOG_READ_RANGE 1023 // 10-bit ((2**10)-1); 10-bit is specific to your
microcontroller (see datasheet)
#define ASTABLE_MIN_BLINK_RATE_HZ 5
#define ASTABLE_MAX_BLINK_RATE_HZ 10
#define MONOSTABLE_MIN_DURATION_MS 3000
#define MONOSTABLE_MAX_DURATION_MS 10000


// DEFINE ALL VARIABLES
// volatile keyword used to insure integrity of value when manipulated by an ISR
volatile bool illuminate_led = 0; // 0 = LOW = False (vs. 1 = HIGH = TRUE)
volatile unsigned long illuminate_on_time_ms = 0;
unsigned long current_time_ms = 0;
int astable_pot_voltage_raw;
int monostable_pot_voltage_raw;
// int astable_pot_voltage;
unsigned long led_blink_delay_ms;
unsigned long led_on_duration_ms;
bool read_illumination_state = 1; // read analog voltages after button press to set
blink rate and on time

// DECLARE ALL FUNCTION (not main)
```

```
void do_this_after_button_press(void);

// setup is run once when the device is powered on
void setup() {
// enable digital pins
pinMode(LED_PIN, OUTPUT);
pinMode(INV_LED_PIN, OUTPUT);
pinMode(BUTTON_INTERRUPT_PIN, INPUT_PULLUP); // pull pin up to HIGH if not driven by
input signal
// associate an ISR with a pin event
attachInterrupt(digitalPinToInterrupt(BUTTON_INTERRUPT_PIN),
do_this_after_button_press, FALLING); // FALLING = HIGH -> LOW transition
// start the serial (communication) monitor
// 9600 is the baud rate (rate of data communication, 9600 bits/s)
// set increments; slower is more reliable
Serial.begin(9600);
Serial.println("LOG: Device initialized.");
}

// loop continuously runs after setup until to break/stop it
void loop() {
if (illuminate_led) {
if (read_illumination_state) {
illuminate_on_time_ms = millis();
Serial.println("Button press detected at time (ms):");
Serial.println(illuminate_on_time_ms);

/* Read the astable potentiometer voltage */
astable_pot_voltage_raw= analogRead(ASTABLE_POT_PIN);
Serial.println("DBG: Astable pot voltage:");
Serial.println(astable_pot_voltage_raw);

/* Calculate the blink delay */
// astable_pot_voltage = (float)astable_pot_voltage_raw*HIGH_VOLT/ANALOG_READ_RANGE;
// converting to absolute voltage is not actually necessary
led_blink_delay_ms = 1000/(ASTABLE_MIN_BLINK_RATE_HZ +
(ASTABLE_MAX_BLINK_RATE_HZ-ASTABLE_MIN_BLINK_RATE_HZ)*(float)astable_pot_voltage_raw/A
NALOG_READ_RANGE);
Serial.println("DBG: Set astable LED blink delay to (ms):");
Serial.println(led_blink_delay_ms);

/* Read the astable potentiometer voltage */
```

```
monostable_pot_voltage_raw = analogRead(MONOSTABLE_POT_PIN);
Serial.println("DBG: Monostable pot voltage:");
Serial.println(monostable_pot_voltage_raw);

/* Calculate the on-time */
led_on_duration_ms = MONOSTABLE_MIN_DURATION_MS +
(MONOSTABLE_MAX_DURATION_MS-MONOSTABLE_MIN_DURATION_MS)*(float)monostable_pot_voltage_
raw/ANALOG_READ_RANGE;
Serial.println("DBG: Set monostable LED on time to (ms):");
Serial.println(led_on_duration_ms);
read_illumination_state = 0; // turn OFF to only read this voltages once per button
press
}

/* Toggle the LEDs */
digitalWrite(LED_PIN, HIGH);
digitalWrite(INV_LED_PIN, LOW);
delay(led_blink_delay_ms); // DELAYS ARE NOT THE IDEAL WAY TO DO THIS; WOULD PREFER
TIMERS!
digitalWrite(LED_PIN, LOW);
digitalWrite(INV_LED_PIN, HIGH);
delay(led_blink_delay_ms);
current_time_ms = millis();
// this one-shot implementation stinks; has a jitter of
// LED_BLINK_DELAY_MS x 2; again, use a timer
if ((current_time_ms - illuminate_on_time_ms) > led_on_duration_ms) {
illuminate_led = 0;
read_illumination_state = 1;
Serial.println("One shot duration exceeded at:");
Serial.println(current_time_ms);
}
}
else
{
digitalWrite(LED_PIN, 0);
digitalWrite(INV_LED_PIN, 0);
Serial.println("Bored; nothing to do... going to sleep.");
//sleep_enable(); // dependent on avr/sleep.h
//set_sleep_mode(SLEEP_MODE_PWR_DOWN); // dependent on avr/sleep.h
delay(SLEEP_DELAY_MS); // delay a bit to make sure chip has everything done before
sleeping
//sleep_cpu(); // dependent on avr/sleep.h
```

```
}
}

/* Interrupt Service Routine (ISR)

Code in this ISR must consume minimal resources and run quickly.

*/
void do_this_after_button_press() {
illuminate_led = 1;
}
```

Testing Data:

Monostable and Astable Potentiometer Outputs

Microcontroller Pin 4 (V)	Microcontroller Pin 5 (V)	Time (s)	Frequency (Hz)
1.70	0	2.78	3.75
1.70	0	5.6	2.71
0	0	3.5	2.55
3.5	1.70	5.7	4.7
3.5	0	8.5	3.11
1.70	3.5	8.77	4.8

Monostable and astable outputs were tested shown above; additionally, the frequency created by the light on the arduino was found and a timer was implemented to find the time that the light is on.

Astable 5-10 Hz: → **FAIL**

Monostable: 3-10 seconds → **PASS**

The 95% confidence interval was not met with the astable potentiometer output. This could be due to the fact the oscilloscope may have not been properly functioning or that the power supply for the arduino was not supplying a benchmark number of volts. Moreover, additional factors may include incorrect microcontroller configuration, limitations in the microcontroller's hardware or software capabilities, potentiometer resistance values not aligning with the desired frequency range, misalignment of timing components, external interference, software errors, voltage instability, and inaccuracies in component tolerances. To address this issue, careful verification of microcontroller settings, hardware compatibility, and circuit components is crucial to ensure precise frequency generation in the specified range.

Battery Life and Power Consumption

When idle, the current drawn in the circuit is approximately 25 mA; furthermore, 4 AA batteries can supply around 2500 milliamp hours. All in all, 2500/25 = approximately **100** hours of battery life when the circuit is idle.

Additionally, when the power source is on, the current drawn in the circuit is approximately 38 mA; knowing that 4 AA batteries supply around 2500 milliamp hours, 2500/38 = approximately 65.78 hours of battery life when the circuit is on and LEDs are working.

Pros and Cons of Discrete Electronics vs. Microcontroller

Using a microcontroller in electronics design offers several advantages, including material and space savings, as well as faster implementation of logic, which can be especially beneficial when working under tight deadlines. Designing PCBs can be time-consuming, involving the creation of electronic schematics and fitting components efficiently. Microcontrollers, on the other hand, require less space and streamline the development process. However, there are drawbacks to using microcontrollers. Customization options are limited, making it challenging to adapt the board's shape to enclosure requirements. Delays from code execution can impact testing accuracy and speed. Moreover, limited onboard memory, relatively slower processing speeds, and single-core operation exist with microcontrollers. Nevertheless, microcontrollers excel in power efficiency and flexibility, as their code can be easily modified, unlike discrete electronics. They are suitable for low-power, portable, and cost-effective devices or specific tasks within complex systems. In contrast, discrete electronics may be preferred for intricate functions that cannot be simplified, must operate at high speeds, or need to be cost-effective in bulk production.